

Product Quality and Safety Management in Large-Scale Systems

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Abstract—A meat-processing enterprise as a complex system is considered. The algorithmic and software support for the numerical realization of the “Decision Tree” for each stage of the trophological chain for meat product manufacture from the field to fork, allowing to reveal hazards, identify risks, establish and characterize critical control points, is briefly described.

Keywords—*quality control, traceability, technological process, parametric model*

I. INTRODUCTION

D.A. Tsvirkun [1] gives a definition of large-scale systems: “is defined by the composition and interrelationships of control as well as production and transport elements in the dynamics of their development and functioning, distribution of management functions by elements of the organizational hierarchy and tasks by production and transport elements.”

A meat processing enterprise is a complex system, and quality control is a complex process, which should involve all the stages - from raw meat production (animal breeding) to the retail of finished products. Providing the population with quality and safe food products is possible only with quality management carried out at all stages and levels of the trophological chain. Thus, a meat processing plant can be viewed as a large-scale system.

The process approach is one of the basic principles of the quality management system in accordance with ISO 9000-2015 [2]. Management systems currently used at enterprises can be displayed as a pyramid (see Figure 1).

SCADA systems, which operate the technological process of the product, are at the bottom level.

SCADA-systems (Supervisory Control And Data Acquisition) are program elements that provide the creation of software for the automation of real time control and management of the technological process. The main purpose of the program created with SCADA is to provide the operator, who controls the technological process, with full information about the process and the necessary tools to influence it.

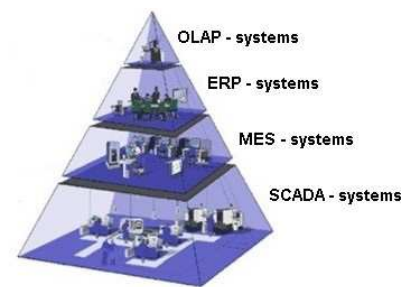


Fig. 1. The pyramid of managerial production systems.

In the production area, HACCP (Hazard Analysis and Critical Control Points) concept must be applied [3]. According to it, the development of the production process system should include three steps:

- assessment of hazards (Hazard) associated with a particular food product and the risk identification (Risk analysis);
- determining critical control points (CCP) where unacceptable risks can occur;
- identification and monitoring of control parameters by which existing hazards can be prevented or reduced to acceptable parameters.

The idea of the quality management system is to minimize nonconformities and ensure traceability of manufactured products by creating an operative system of management decisions and assigning roles and responsibilities to avoid chaos. The quality management system is not a panacea for all kinds of problems; it is only a tool to achieve maximum results with available resources.

II. TRACEABILITY AND INTEGRATED CONTROLS

Ensuring the food product traceability is becoming the key in solving problems of national importance, such as sustainable quality of food products [4], ensuring food safety [5], ensuring that the consumer [6] is informed about the composition and origin of food ingredients.

In Russia, traceability is primarily associated with the increased use of FSIS Mercury [7]. Maintaining the internal electronic certification of all types of cargo, tracking the path of their shipment through Russia, and improving biological and food safety is the main function of the Mercury system. However, at the moment, FSIS Mercury provides for low-resolution traceability, which allows determining, for example, what farm milk was used to make a batch of products sold on a specific date. FSIS Mercury does not record operations of the production process. The production process is presented as a “black box,” for which only input raw materials and output products are recorded. This situation has technical solutions, but it requires quite serious changes to the whole system for implementation.

Open communication and the use of appropriate information technology are required for the concept of

integrated control. Everyone involved in production (from feed producers to retailers) must keep appropriate activity records, and the relevant data must be made available to the authorities. This is the only way to achieve traceability of a product. Such traceability is an important part of the concept of integrated control. It requires control of animal feeding, identification of animals and means of their transportation, keeping records of antibiotic use and other treatment of animals, clear labeling of each batch of animal products, and an indication of all food elements at sale [8].

Specialists of the V.M. Gorbатов All-Russian Meat Research Institute (since 2017, the V.M. Gorbатов Federal Research Center for Food Systems of the Russian Academy of Sciences) proposed the following traceability chart (see Figure 2).

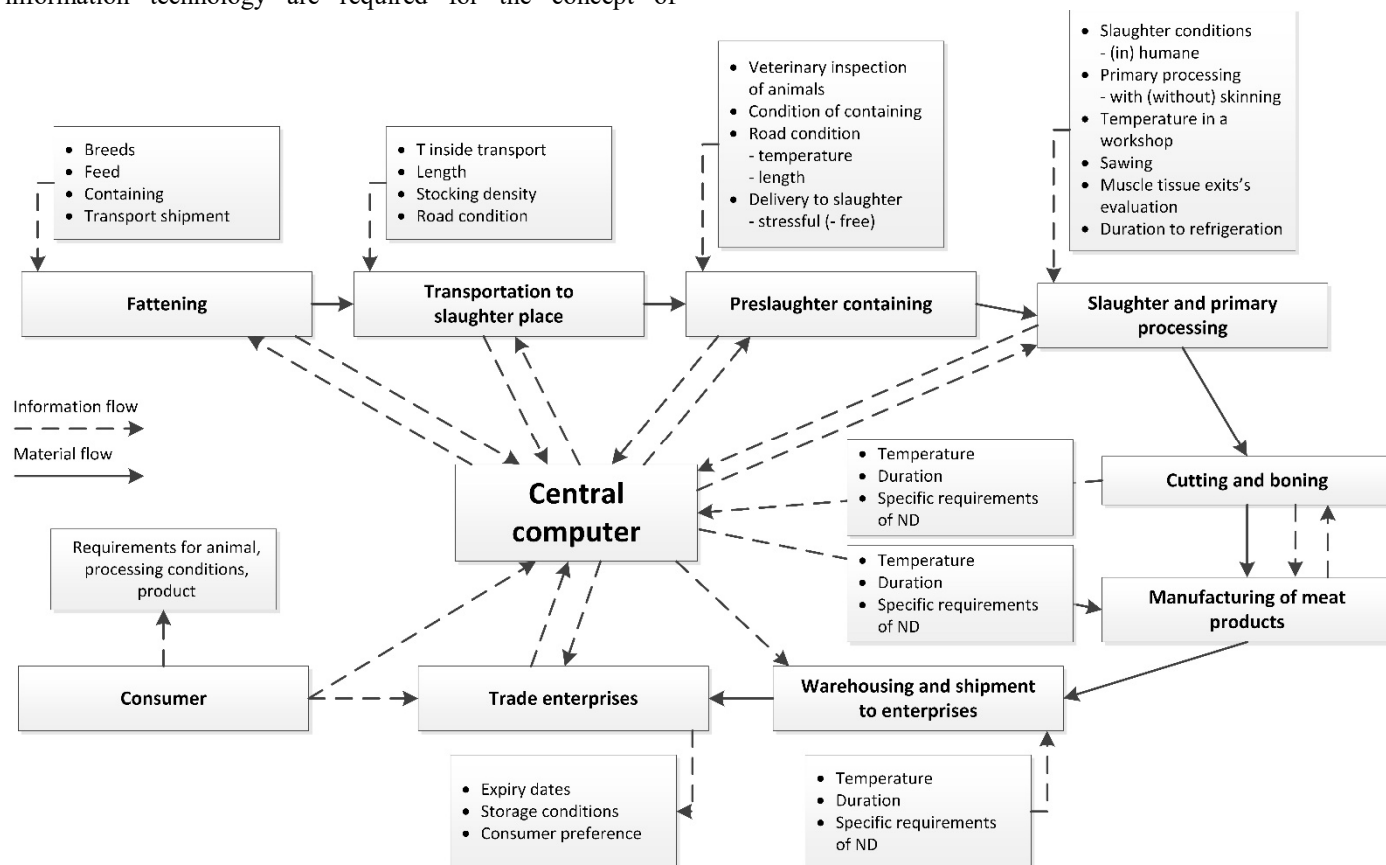


Fig. 2. Traceability for the meat production chain from field to fork.

III. COMPREHENSIVE SYSTEM OF QUALITY AND SAFETY ASSURANCE

A practical solution to the problem of providing consumers with quality-assured products, preventing food contamination, and ensuring food safety could be the implementation of the concept of producing safe quality-assured products through the formation of a single trophological chain “from field to market,” the essence of which is an interrelated sequence of individual links in the chain: feed production – growing – transportation to slaughter – slaughtering – processing –

production of finished products – storage and transportation – sale – consumption [8].

This requires an analysis of all stages of the production process, taking into account the risks that are classified as unacceptable - a high and medium risk zone. Additionally, the impact of subsequent stages of the production process on the likelihood of risks being realized must also be taken into account. In works [8, 9], it is noted that in the meat production process, 69 % of all identified hazards are biological, 21.6 % - physical, and 9.4 % - chemical. In general, CCPs are identified at the following stages: 1) feeding; 2) dehiding; 3) deboning; 4)

control of the active ingredient in the finished product; 5) storage at the retail; 6) storage at the consumers.

To justify control points, the food safety management system is based on the following principles:

- hazard analysis and risk identification at all stages of production;
- determination of CCPs;
- establishment of critical limits for each CCP - defining the criterion that indicates that the process is under control;
- establishment of procedures for CCT monitoring;
- developing corrective actions if the process is out of control;
- recording and implementing documentation;
- verifying the feasibility of the system.

The first stage is the decomposition of the technological process to the level of technological operations

$$TP = \{TO_i(CCP), TO_i, H_i\}, i = \overline{1, N}, \quad (1)$$

where TP – technological process; TO – technological operation; $TO_i(CCP)$ – the i -th technological operation, in which a control action can be applied, i.e., there is a critical control point; TO_i – the i -th technological operation, to which a control action cannot be applied, i.e., there is no critical control point; H_i – weight factor.

At the second stage, a parametric model of the technological process is formed. The parametric model takes into account 1) sequence of technological operations, 2) set of parameters by means of which traceability, controllability, and manageability of both separate technological operations and the whole technological process are provided, 3) ranges of values of each of the parameters, controlled object, etc. [10-13].

The parametric model in vector-multiple representation can be represented as:

$$P = \{P_i, N\}, i = \overline{1, N} \quad (2)$$

$$P_i = \{PR_{ij}, \alpha_{ij}, CCP_{ij}, \beta_{ij}\}, j = \overline{1, M} \quad (3)$$

$$PR_{ij} = \{PR_{ij}^T, PR_{ij}^*, nor\{PR_{ij}\}, min\{PR_{ij}\}, max\{PR_{ij}\}, \tau\} \quad (4)$$

$$CCP_{ij} = \{CCP_{ij}^T, CCP_{ij}^*, nor\{CCP_{ij}\}, min\{CCP_{ij}\}, max\{CCP_{ij}\}, \tau\}, \quad (5)$$

where $i = \overline{1, N}$ – vector of the number of technological operations; $j = \overline{1, M}$ – vector of the number of j parameters at the i -th technological operation; P – vector of parameters controlling the production process; P_i – vector of parameters of the i -th technological operation; PR_{ij} – vector of the j -th

control point at the i -th technological operation; CCP_{ij} – vector of the j -th CCP at the i -th technological operation; α_{ij}, β_{ij} – dimensionless weight factors, respectively, for each component of vectors P_i and CCP_{ij} ; $PR_{ij}^T, PR_{ij}^*, nor\{PR_{ij}\}, min\{PR_{ij}\}, max\{PR_{ij}\}$ – current, set, normalized, minimum and maximum values of the components of the vector PR_{ij} ; $CCP_{ij}^T, CCP_{ij}^*, nor\{CCP_{ij}\}, min\{CCP_{ij}\}, max\{CCP_{ij}\}$ – current, set, normalized, minimum and maximum values of the components of the vector CCP_{ij} ; τ – time of the current coordinate.

Parametric models are built for all process links from the point of view of the process approach (Table 1).

TABLE I. PARAMETRIC DESCRIPTIONS ALL TECHNOLOGICAL PROCESS LINKS

Input	Control (controllable/uncontrollable parameters)	Output
SLAUGHTER AND PRIMARY PROCESSING		
Ready for slaughter farm animal	Transfer of microorganisms with water; Contamination of carcasses with microorganisms when skinning and evisceration; Appropriate bleeding; Disinfectants, lubricant oils from equipment and inventory; «Fortificants» (vitamins, minerals and etc.); Transfer of shards from inventory or remains of bone tissue to meat; Noncompliance with temperature and humidity regimes, consequently, the growth of microorganisms	Carcass Meat cuts Meat Byproducts
CUTTING		
Carcass, Meat cuts, Meat	Environment parameters: - temperature; - humidity	Ready for processing raw material
MINCED MEAT PREPARATION		
Raw material	Environment parameters: - temperature; - humidity. Minced meat temperature; Order of ingredient adding; Noncompliance with temperature and humidity regimes; Overdose of nitrates and phosphates; Contamination with washing disinfectants and lubricants; The appearance of foreign particles to products; Noncompliance with the rule «first in - first out»; Cross-contamination of products by personnel	Minced meat
HEAT TREATMENT AND REFRIGERATION		
Minced meat	Noncompliance with temperature and humidity regimes, consequently, growth of microorganisms; Formation of toxic compounds (PAH, HCA, etc.) during the smoking process	Ready for packing products
PACKING, LABELING AND KEEP		
Ready for packing products	The growth of microorganisms when a package is broken; Noncompliance with temperature and humidity regimes; Transfer of foreign particles to a ready product;	Ready product

	Noncompliance with the rule «first in - first out»; Cross-contamination of products by staff.	
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According to the general principles of HACCP and using “Decision Tree,” critical control points of the trophological chain of meat products production from field to consumer are defined. In determining these, a computer program developed to implement the “Decision Tree” was used.

In the dialogue mode, the required operation is selected. For this operation, the evaluated hazard factor is selected, and further, it is required to answer the questions of the “Decision Tree”. According to the answer choice, the presence or absence of a critical control point is automatically determined (see Figure 3).

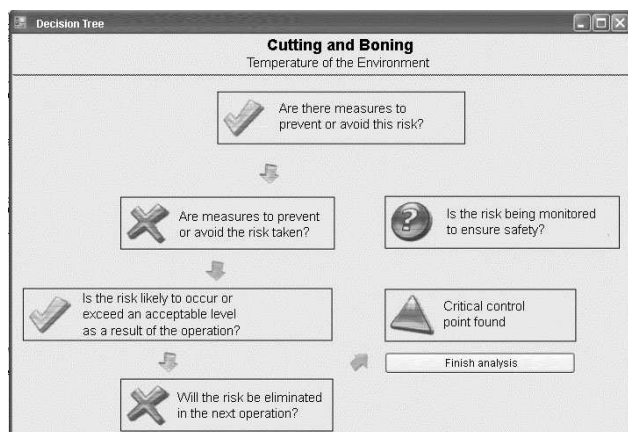


Fig. 3. Interface “Decision Tree”.

Then for each critical control point, a critical limit – the value that separates the acceptable level from the unacceptable one – must be set. Critical limits are set by the HACCP working group for each CCP for one or more parameters. Sources of information include published scientific data, experimental results, regulations, expert recommendations, mathematical models, etc. Critical limits should be specified and confirmed. Confirmation of critical limits is proof that the selected critical limit actually controls the hazardous factor [14].

A monitoring scheme, i.e., a system of continuous observations or measurements, is automatically developed for each critical point to verify that the critical points are under control. The software automatically generates monitoring logs of the CCPs. All HACCP data and documents can be generated and stored electronically [15].

IV. CONCLUSIONS

A meat processing plant is considered a large scale system. The quality management system occupies a crucial place in the management of the enterprise. Solving the tasks of traceability and predictability will improve the quality of products. The automated information system Mercury is in operation in Russia for the electronic certification and traceability of goods subject to state veterinary oversight during their production, circulation, and movement through the Russian Federation in

order to create a unified information environment for veterinary medicine and improve biological and food safety. However, in FSIS Mercury, the production process is a “black box.” It is also worth noting that traceability and forecasting tasks in a meat processing plant are complicated by the multi-factor nature of the models and the polyvariety of characteristics. Thus, the use of simulation modeling in solving these problems will allow implementing operative management of production processes and optimization of technological parameters in real time mode. The software for meat production quality and safety control, based on HACCP principles, is presented. It can be used as a training program for experts and HACCP auditors and promotes quality control system implementation in the meat industry.

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